Datasheet

ESP32 MicroPLC

Part No.: MicroPLC-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Relay Board)

Controller: ESP32-WROOM-32U



Functional Overview

The **HomeMaster MicroPLC** is a compact, open-source DIN rail controller built around the ESP32 platform, designed for reliable automation in residential, commercial, and light industrial applications. Despite its small footprint, it integrates essential industrial interfaces including a digital input, relay output, RS-485 communication, real-time clock, and USB programming interface.

Engineered for simplicity and reliability, the MicroPLC operates from a **24 V DC supply** and provides an integrated **RS-485 Modbus interface** for communication with field devices and expansion modules. Local processing on the ESP32 ensures stable operation even without cloud connectivity, enabling dependable edge control for distributed automation systems.

two hardware boards (MCU board, relay board).

Board	Function
MCU Board	CPU, USB
Relay Board	Power supply, relays, digital inputs

Technical Specifications

Specification	Details
Microcontroller	ESP32-WROOM-32U-N16 (dual-core)
Power Input	24 V DC nominal
RTC	RTC (PCF8563); backup battery holder, battery not fitted
Document revision	DS-MicroPLC Rev. B · 2026-09 · Hardware V1.0
Digital Inputs	1 × IEC 61131-2 compliant 24 V digital input (ISO1212 front-end) (DIx + GNDx), dry-contact (module-wetted), not galvanically isolated; per-channel PTC fuse, TVS surge suppression, reverse-polarity protection, EMI filtering
Relay outputs	1 × relay, C and NC only (SPST-NC); 3 A @ 250 VAC (module limit)
Temperature Inputs	1 × 1-Wire (DS18B20 compatible)
User Interface	1 front-panel button, 6 LEDs (1 user, 1 status, RX, TX, relay, DI)
RS-485	half-duplex Modbus RTU, not galvanically isolated
Wi-Fi	Wi-Fi (ESP32)
USB	USB-C (ESD protected, CC detection, data to ESP32)

Top Terminal Block (1–6)

Pos	Marking	Signal	Description
1	+V	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	I.1	DI1	DI1 input
4	GND	DI1	DI1 return
5	C	RELAY	Relay common
6	NC	RELAY	Relay normally closed

Bottom Terminal Block (1–6)

Pos	Marking	Signal	Description
1	COM	RS485	RS-485 signal reference
2	B	RS485	RS-485 data -
3	A	RS485	RS-485 data +
4	+5V	BUS	1-Wire sensor supply
5	D	BUS	1-Wire data
6	Gnd	BUS	1-Wire ground

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses

or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Installation, Environmental & Mechanical

Category	Specification	Details
Environmental Ratings	Operating temperature	0 °C to +40 °C
Environmental Ratings	Storage temperature	−10 °C to +55 °C
Environmental Ratings	Relative humidity	0–90 % RH, non-condensing
Environmental Ratings	Ingress protection	IP20 (inside cabinet only)
Environmental Ratings	Installation	Indoor control cabinet only; not for outdoor or exposed installation
Environmental Ratings	Maximum altitude	2000 m
Environmental Ratings	Pollution degree	2
Mechanical & Packaging	Dimensions	35.5 × 90.6 × 67.3 mm (L × W × H)
Mechanical & Packaging	DIN width	2 modules (≈ 35.5 mm)
Mechanical & Packaging	Mounting	35 mm DIN rail
Mechanical & Packaging	Enclosure	PC/ABS industrial enclosure
Mechanical & Packaging	Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Mechanical & Packaging	Wire cross-section	0.2–2.5 mm^2 (AWG 24–12)
Mechanical & Packaging	Tightening torque	0.4–0.6 Nm
Mechanical & Packaging	Net weight	120 g
Mechanical & Packaging	Gross weight	220 g
Mechanical & Packaging	Pack size	TBD

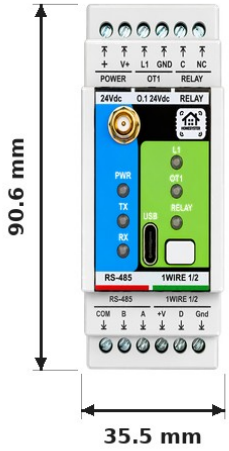
Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.

MicroPLC

All dimensions in mm

Front view



90.6 mm

35.5 mm

Side view



67.3 mm

Cable Recommendations & Shield Grounding

This section applies to **Temperature (1-Wire)**, and **RS-485**. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables (1-Wire/RS-485) separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

Temperature Cable

1-Wire (DS18B20)

- **Recommended:** shielded 3-core (+5V / DATA / GND) for typical installations
- **High-EMI / long runs:** shielded pairs + overall shield (e.g. **LI2YCY PIMF 2×2×0.50**)
- **Topology:** daisy-chain (bus). Avoid star wiring.
- **Stubs:** keep sensor stubs ≤ **0.5 m**.
- **Pull-up (DATA): 4.7 kΩ** typical; **2.2–3.3 kΩ** for long/heavy loads.

RS-485 Cable

- **Construction:** twisted pair for **A/B**
- **Characteristic impedance: 120 Ω** (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples: J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**
- Use one twisted pair for A/B. Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Shield Grounding

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (1-Wire/RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

Compliance & Certifications

The HomeMaster MicroPLC is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

EU Directives

EMC 2014/30/EU · LVD 2014/35/EU · RED 2014/53/EU · RoHS 2011/65/EU

Harmonised Standards

Area	Standards
EMC	EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions)
Electrical Safety	EN 62368-1
Radio	EN 300 328 · EN 301 489-1 · EN 301 489-17
RoHS	EN IEC 63000

Radio

The product integrates a pre-certified ESP32 Wi-Fi radio module (2.4 GHz). Final product conformity with the Radio Equipment Directive is demonstrated by the maintained technical documentation and conformity assessment of the complete device.

Safety notice

24 V DC: SELV · Qualified personnel only

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